

Design Number: 1 & 2

Design Name: Variable Gain Signal Conditioning Circuits

Station Number: 1

Submission Date: October 28, 2025

Objective

The objective of this experiment was to observe and study the effects of variable gain conditioning circuits on input signals through use of the LM358 op-amp and PSpice simulation.

Equipment

- (1) Computer with PSpice software
- (1) Breadboard
- (1) LM358 IC
- Jumper Wires
- (1) Variable Resistor Decade Box
- (1) Signal Generator with Appropriate Cables
- (1) Oscilloscope with Appropriate Cables
- (1) DC Power Supply

Procedure

Simulation:

1. Open ORCAD's PSpice software by searching "Capture" in the Windows menu, and make a new file.
2. Construct the circuit of figure 1 below by dropping parts and connecting with wires.

Note: you must update the parts library to find the necessary op-amp IC.:

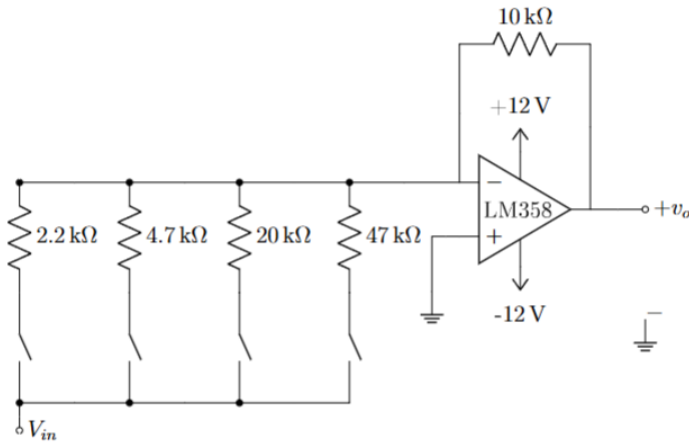


Figure 1: The following circuit is a poorly designed VGA circuit which we will later improve upon.

3. Choose one of the four resistance values. Apply an input voltage signal as a sinusoid with a 200 mV peak-to-peak value and a frequency of 1 kHz. (*Hint: the voltage source should be “Vsin”)
4. Make a new simulation profile, and run the simulation.
5. Measure and record the output voltage.
6. Add a new trace to the waveform. Apply the db function to the output voltage. Observe the output response compared to the input signal.
7. Repeat steps 3-6 for the remaining three resistance values and record all values by completing table 1 below:

R_f	R_i	V_o	V_i	$\frac{R_f}{R_i}$	$\frac{V_o}{V_i}$	dB	Error %

Table 1: The following table will be used to record gain responses with resistance changes.

8. Reconstruct the circuit from before to reflect that of figure 2:

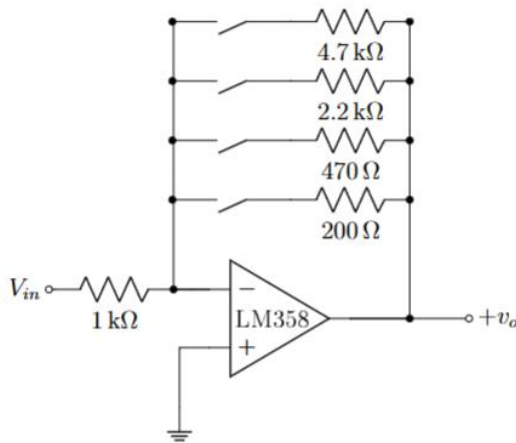


Figure 2: The following is an improved iteration of the VGA circuit from figure 1.

9. Assign the value of R_i to $1\text{ k}\Omega$ as reflected in figure 2, and select one of the four resistance values for R_f .
10. Repeat steps 3-6 for the remaining three resistance values and record all values by completing a new iteration of table 1 noting its distinction from the previously completed table.

Physical Construction:

1. Obtain the breadboard, IC, resistor box, and function generator, and construct the circuit of figure 1.
2. Set the function generator to output a sinusoid signal with a peak-to-peak voltage of 200 mV and a frequency of 1 kHz.
3. Connect assure the DC power supply is properly limited in its current before applying 12 V to the positive and negative rails of the op-amp IC.
4. Connect the oscilloscope to the output of the op-amp ensuring proper grounding.
5. Choose one of the four resistance values and apply that to the resistor box.
6. Select the “Auto Range” button on the oscilloscope, and observe and record the output.
7. Repeat steps 5 and 6 for each of the remaining resistance values. *Note: You must record the output voltages for each iteration to calculate the percent error for both entries of table 1.
8. Rearrange the circuit to reflect the improved circuit of figure 2.
9. Repeat steps 2-8 for the new circuit topology.
10. Note any differences and explain results.

Results

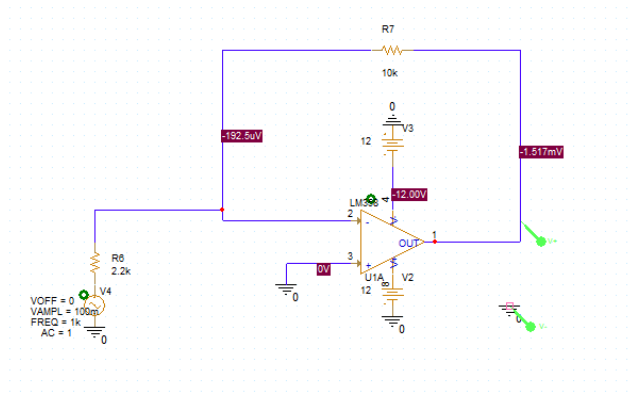


Figure 3: The following is the simulated circuit of figure 1 when R_i is $2.2\text{ k}\Omega$.

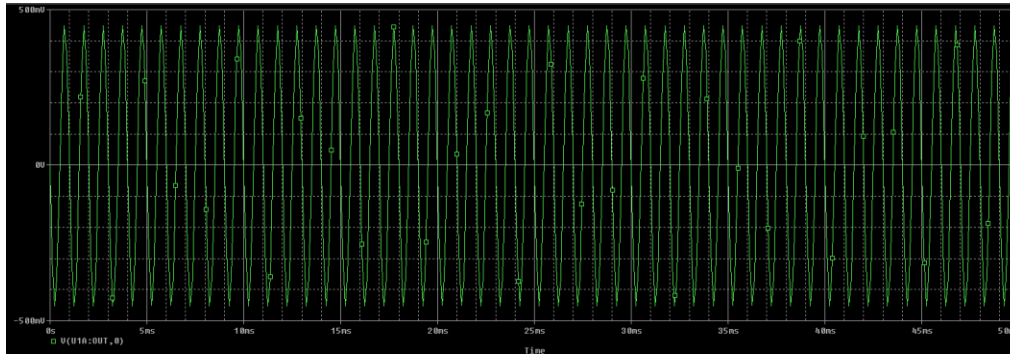


Figure 4: The following is the waveform for figure 3.

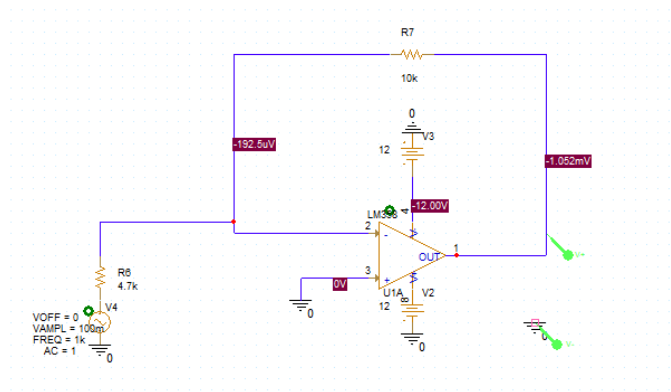


Figure 5: The following is the simulated circuit of figure 1 when R_i is $4.7\text{ k}\Omega$.

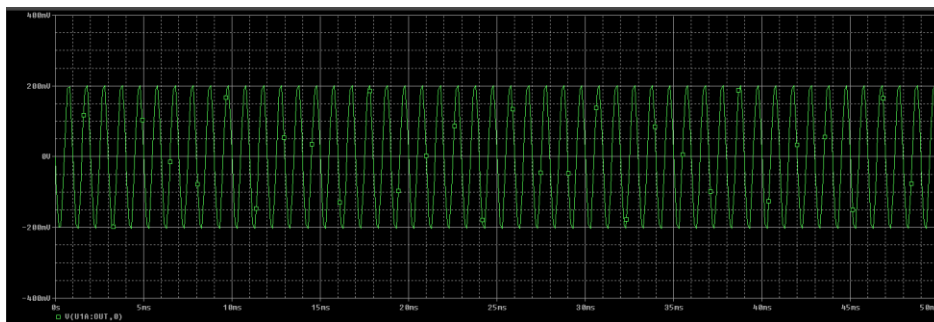


Figure 6: The following is the waveform for figure 5.

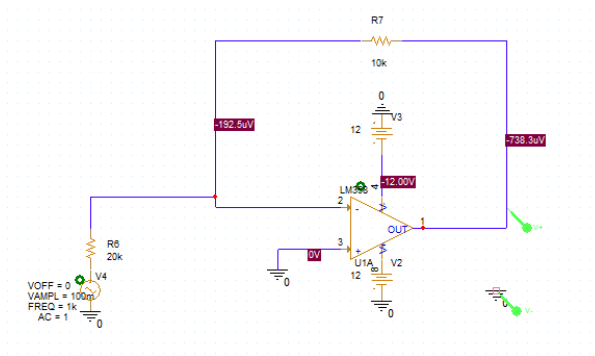


Figure 7: The following is the simulated circuit of figure 1 when R_i is 20 k Ω .

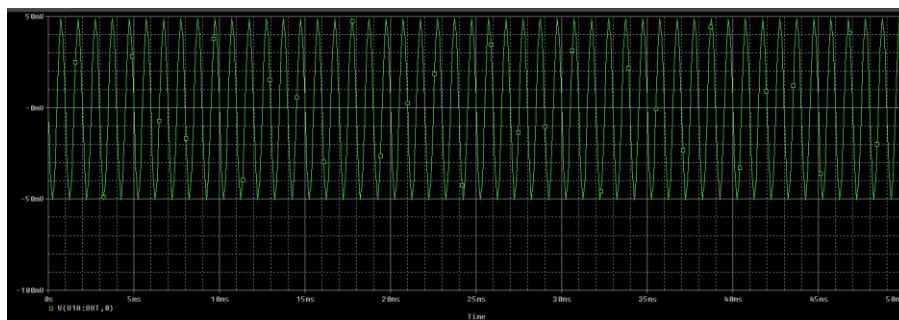


Figure 8: The following is the waveform for figure 7.

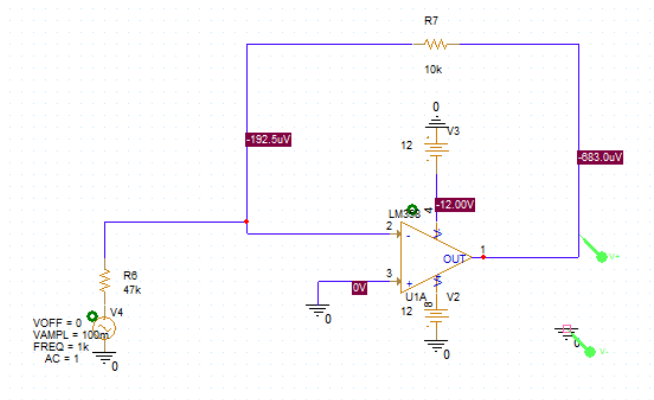


Figure 9: The following is the simulated circuit of figure 1 when R_i is 47 k Ω .

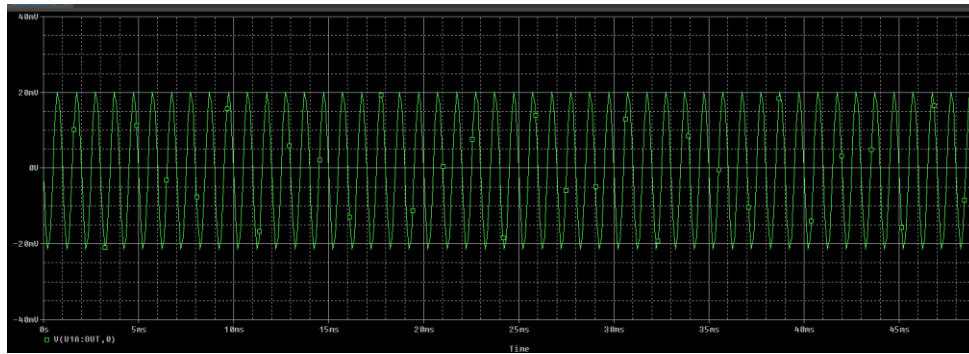


Figure 10: The following is the waveform for figure 9.

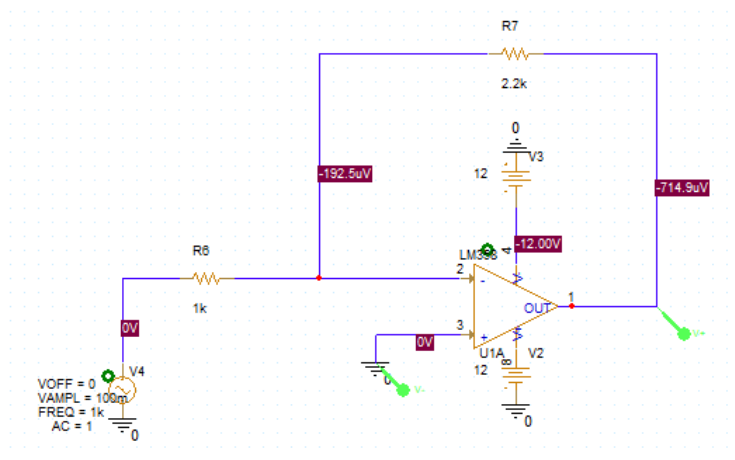


Figure 11: The following is the simulated circuit of figure 2 when R_f is $2.2\text{ k}\Omega$.

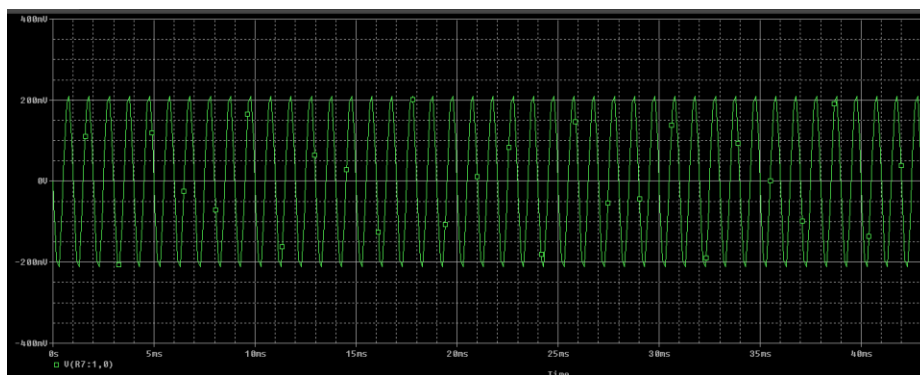


Figure 12: The following is the waveform for figure 11.

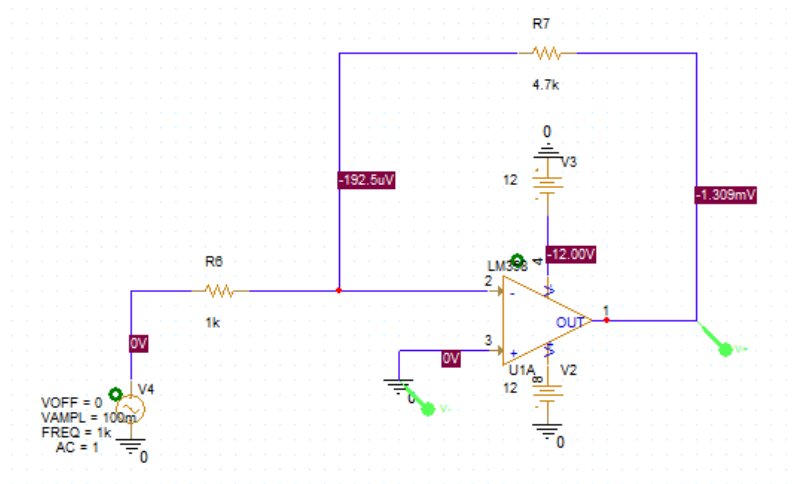


Figure 13: The following is the simulated circuit of figure 2 when R_f is 4.7 k Ω .

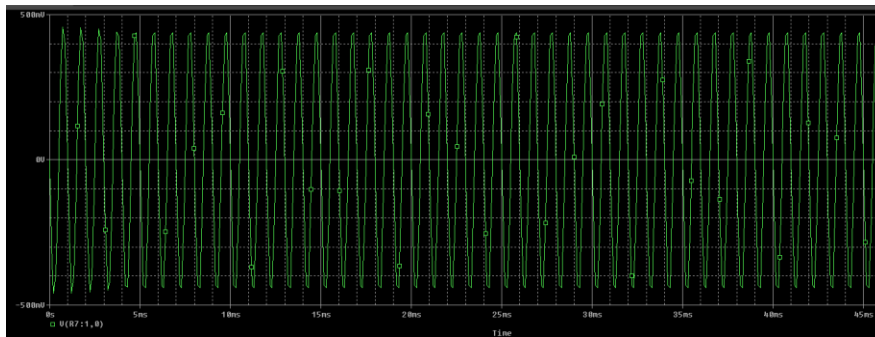


Figure 14: The following is the waveform for figure 13.

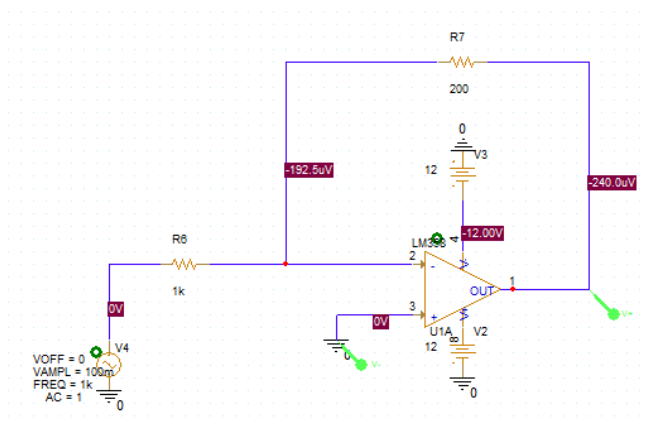


Figure 15: The following is the simulated circuit of figure 2 when R_f is 200 Ω .

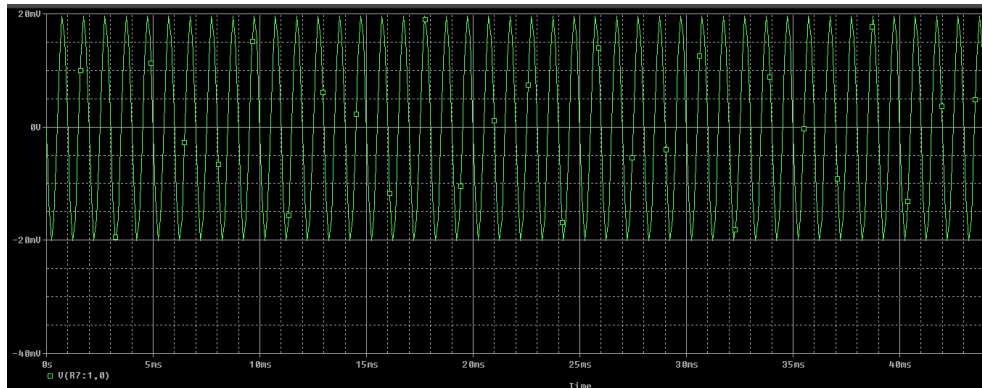


Figure 16: The following is the waveform for figure 15.

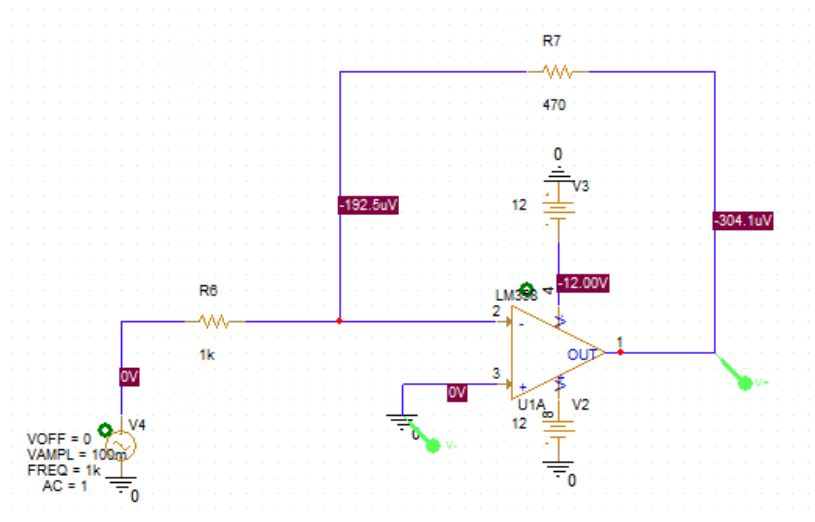


Figure 17: The following is the simulated circuit of figure 2 when R_f is $470\ \Omega$.

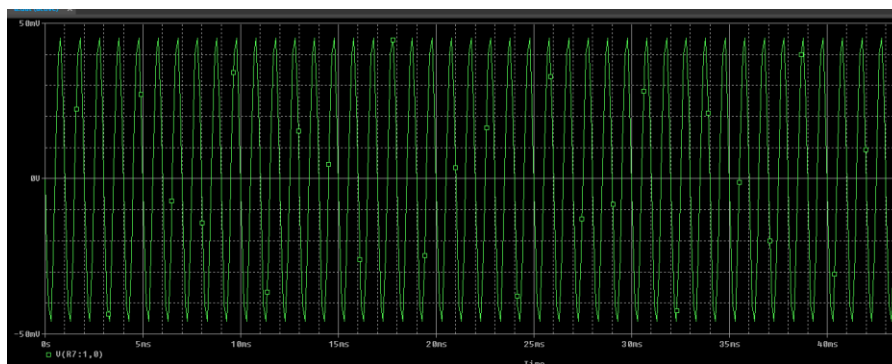


Figure 18: The following is the waveform for figure 17.

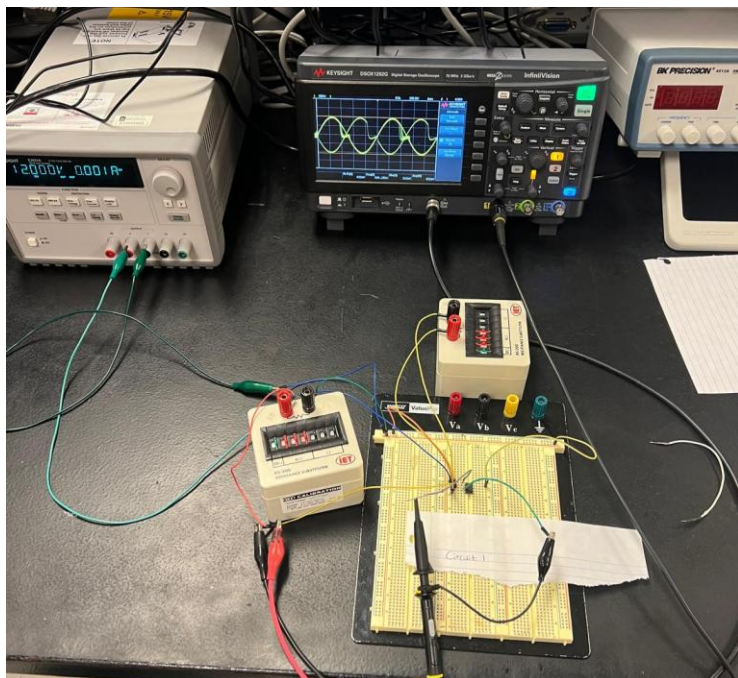


Figure 19: The following is physical construction of circuit 1.

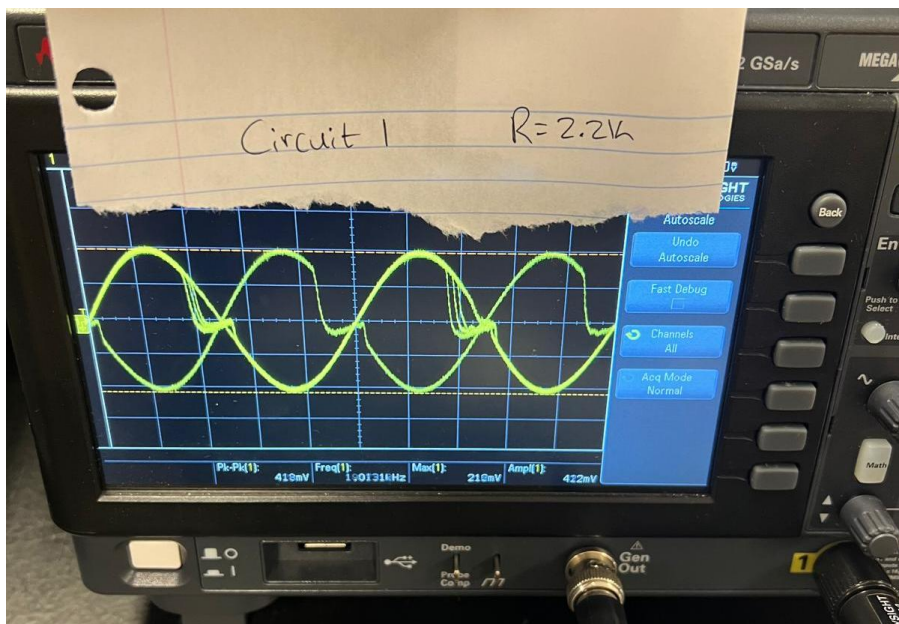


Figure 20: The following is waveform of the physical construction of figure 1 when $R = 2.2 \text{ k}\Omega$.

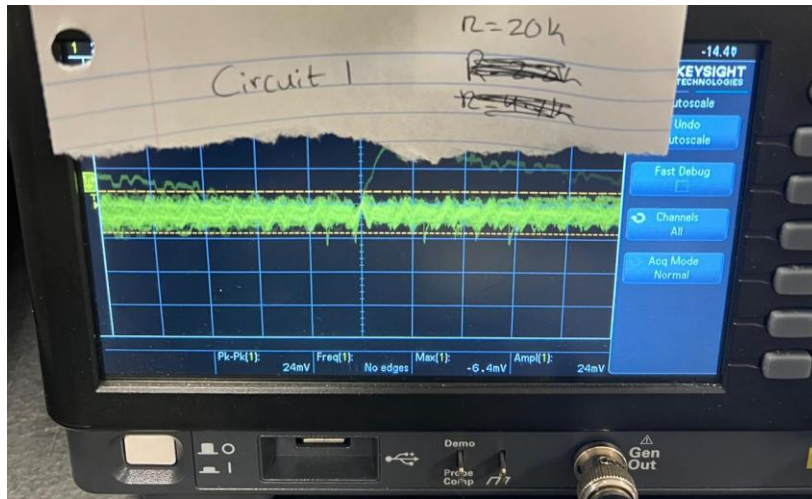


Figure 21: The following is waveform of the physical construction of figure 1 when $R = 20\text{ k}\Omega$.

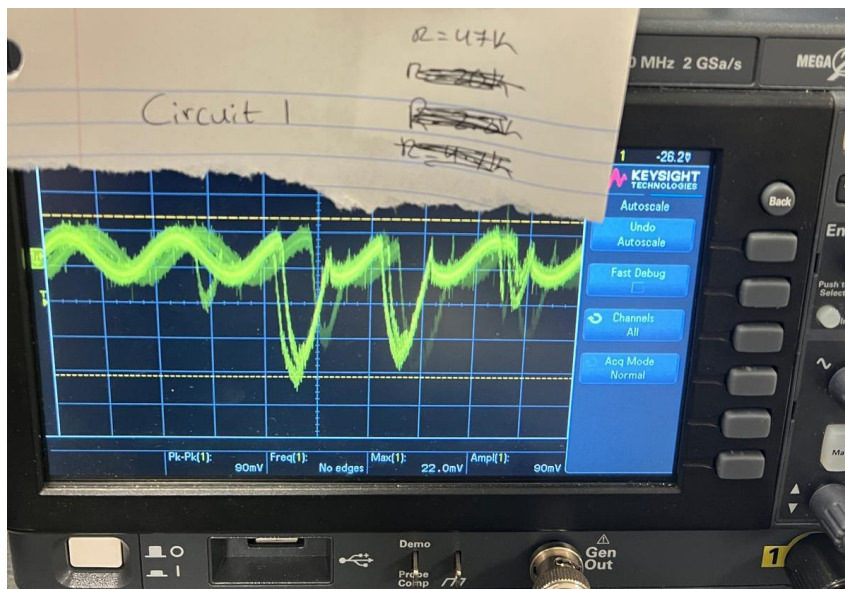


Figure 22: The following is waveform of the physical construction of figure 1 when $R = 47\text{ k}\Omega$.

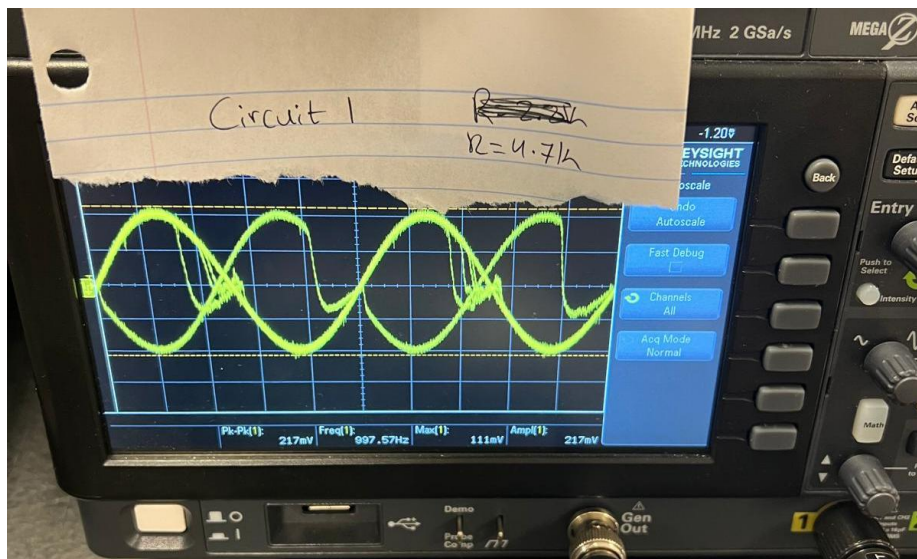


Figure 23: The following is waveform of the physical construction of figure 1 when $R = 4.7 \text{ k}\Omega$.

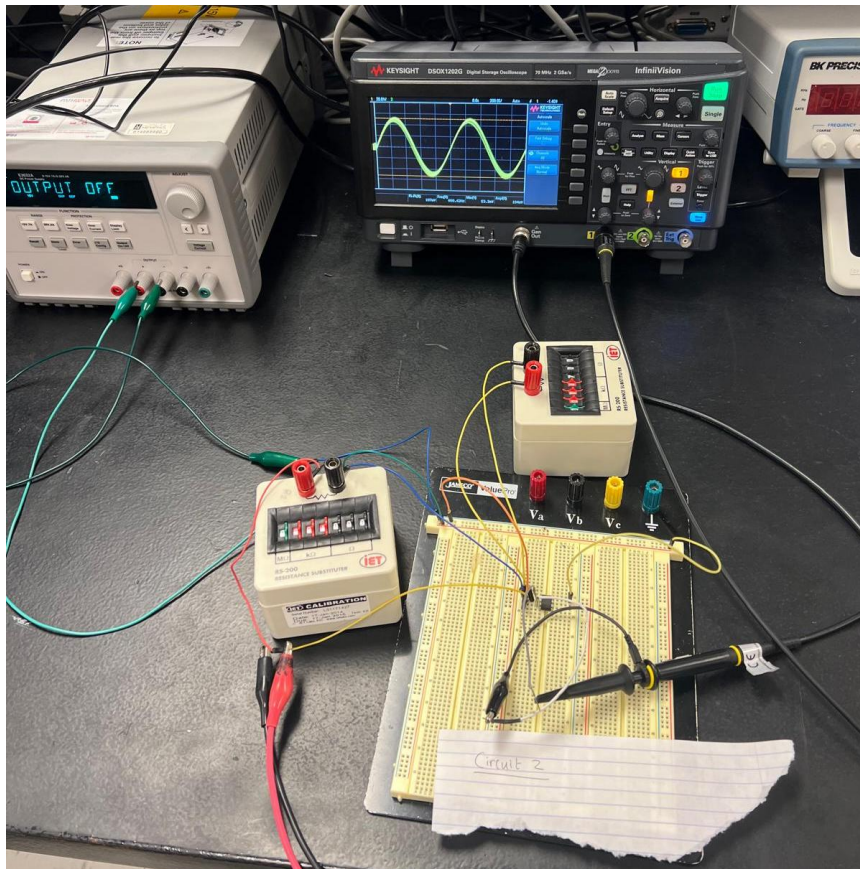


Figure 24: The following is physical construction of circuit 2.

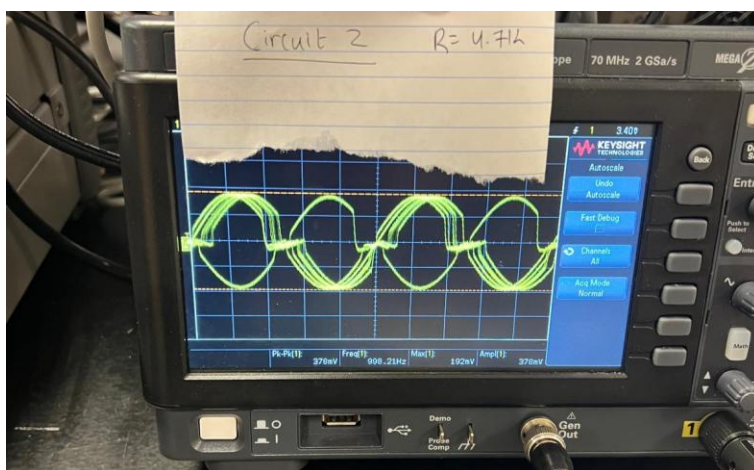


Figure 25: The following is waveform of the physical construction of figure 2 when $R = 4.7 \text{ k}\Omega$.

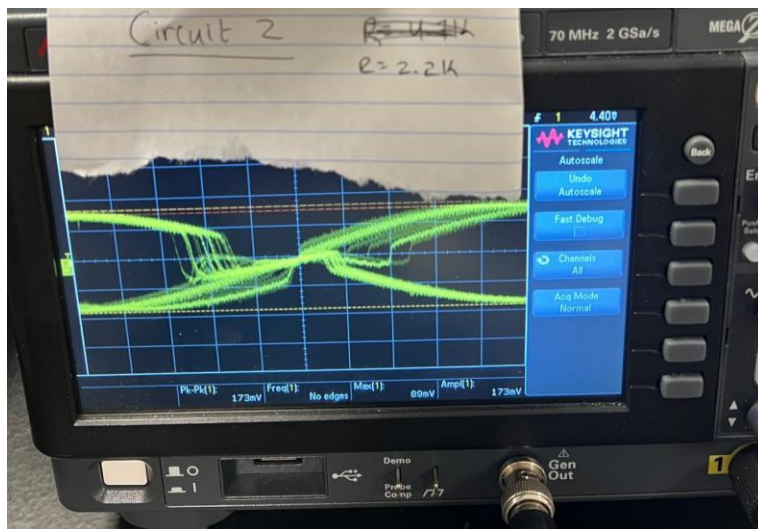


Figure 26: The following is waveform of the physical construction of figure 1 when $R = 2.2 \text{ k}\Omega$.

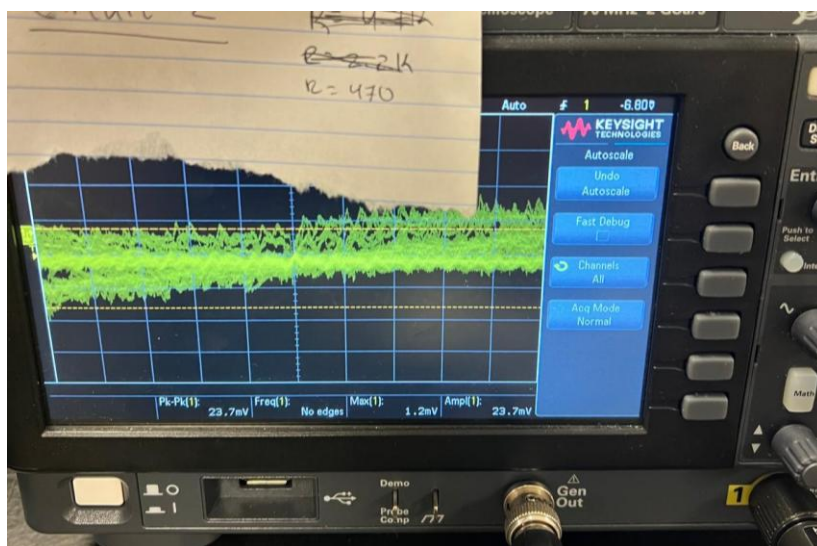


Figure 27: The following is waveform of the physical construction of figure 1 when $R = 470 \Omega$.

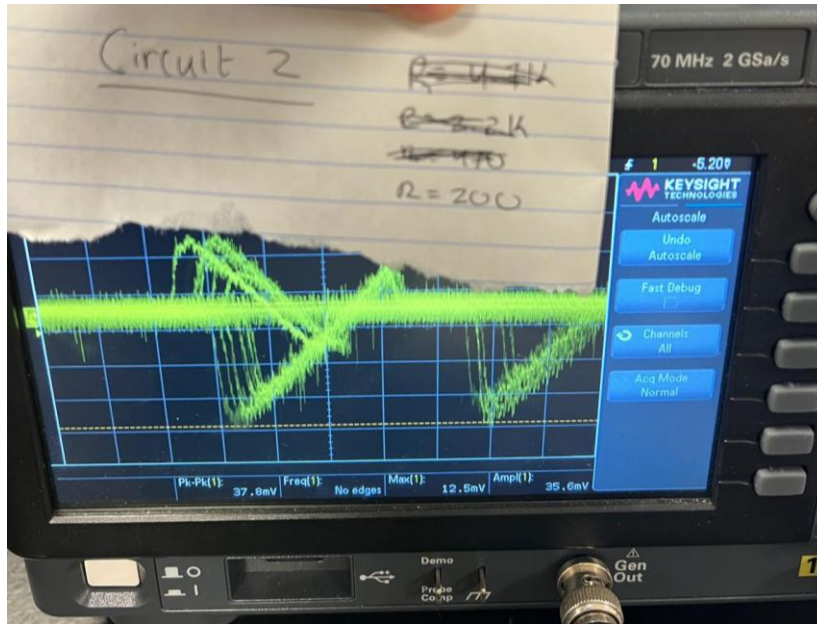


Figure 28: The following is waveform of the physical construction of figure 1 when $R = 200 \Omega$.

R_f	R_i	V_o	V_i	$\frac{R_f}{R_i}$	$\frac{V_o}{V_i}$	dB	Error %
10 k Ω	2.2 k Ω	450 mV	200 mV	4.55	2.25	7.04	7.1
10 k Ω	4.7 k Ω	200 mV	200 mV	2.13	1	0	8.5
10 k Ω	20 k Ω	50 mV	200 mV	0.5	0.25	-12.04	52
10 k Ω	47 k Ω	20 mV	200 mV	0.21	0.1	-20	22.2

Table 1a: The following table represents the results for the configuration of circuit 1. The percent error is calculated by comparing the simulated results to the actual results.

R_f	R_i	V_o	V_i	$\frac{R_f}{R_i}$	$\frac{V_o}{V_i}$	dB	Error %
2.2 k Ω	1 k Ω	200 mV	200 mV	2.2	1	0	13.5
4.7 k Ω	1 k Ω	450 mV	200 mV	4.7	2.25	7.04	16
200 Ω	1 k Ω	20 mV	200 mV	200	0.1	-20	89
470 Ω	1 k Ω	45 mV	200 mV	470	0.225	-12.95	1.2

Table 1b: The following table represents the results for the configuration of circuit 2. The percent error is calculated by comparing the simulated results to the actual results.

Report Questions:

1. Plot the percent error vs gain from the data collected in the procedure.

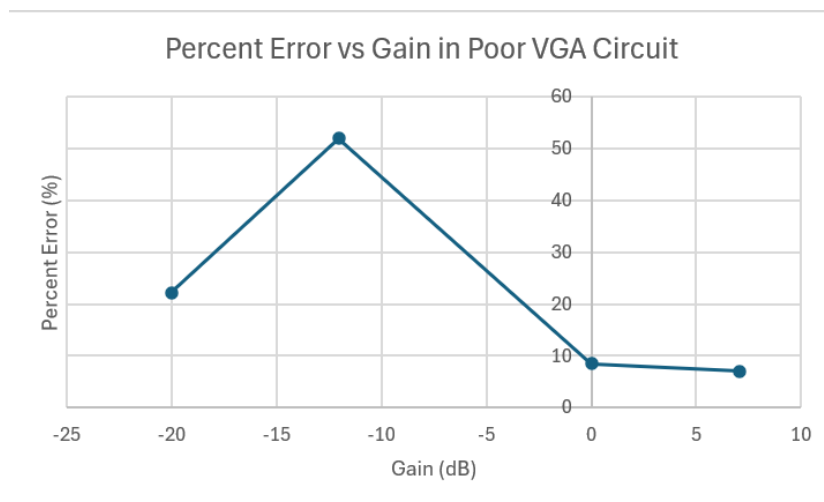


Figure 29: The following graph shows the relationship between percent error and gain in the poorly designed circuit of figure 1.

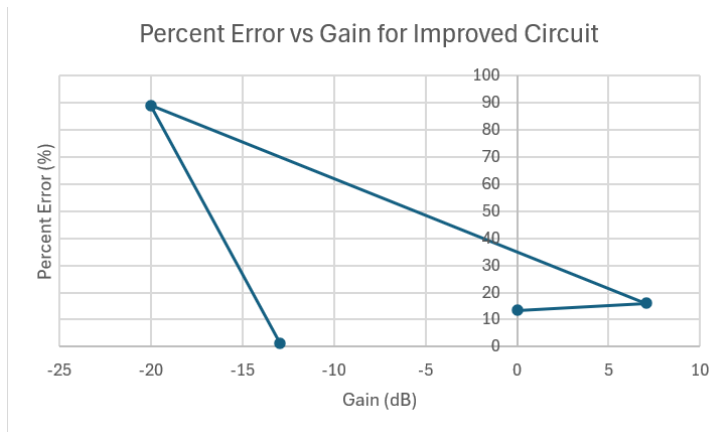


Figure 30: The following graph shows the relationship between percent error and gain in the improved circuit of figure 2.

****Please note that our graphs are not correct due to faulty equipment (oscilloscope) which struggled to display the output voltage in the physical circuit. This was out of our control. *****

Conclusion

Performing this experiment upheld our previous knowledge of amplifier circuits in a practical way. We noticed the effect of changing R_f or R_i values on the gain of an op-amp circuit. We were expecting to notice an improvement in gain conditioning in the second circuit which utilized varying resistance values across the R_f . This was the case in simulation. However, we were not able to fully prove this as our lab equipment was proven to be faulty. We received a lot larger error than acceptable due to a faulty oscilloscope which was confirmed by our instructor. This was especially true for the cases where the measured output voltage was very low (below the 50mV threshold). However, we were able to confirm that this was upheld for higher voltage outputs.

Preparation Problems

No preparation problems were provided for this design lab.

Worksheet

No worksheet is applicable for this experiment as no physical calculations were required for the preparation or report questions.